

Trigonometric Equation Solving

Algebra II / Precalculus Practice

Name: _____

Date: _____

Instructions

Complete all 12 problems in 45 minutes without a calculator. Show the algebra that produces each trigonometric equation and give exact answers. In Problems 1–6, solve on $0 \leq x < 2\pi$. In Problems 7–12, give all real solutions using $k \in \mathbb{Z}$. If an angle is not a standard unit-circle value, leave it in inverse-trigonometric form.

Use only factoring, inverse trigonometric functions, the complementary-angle rules, and the identities

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x.$$

Do not use sum, difference, double-angle, half-angle, or product-to-sum identities. Expressions such as $\cos(2x)$ should be treated as cosine with argument $2x$.

For equations with matching trigonometric functions, the following general patterns may be used:

$$\sin A = \sin B \implies A = B + 2\pi k \quad \text{or} \quad A = \pi - B + 2\pi k,$$

$$\cos A = \cos B \implies A = 2\pi k \pm B,$$

$$\tan A = \tan B \implies A = B + \pi k.$$

Solve on $0 \leq x < 2\pi$

1.

$$2 \sin x + \sqrt{2} = 0$$

2.

$$2 \cos^2 x - 3 \cos x + 1 = 0$$

3.

$$\tan^2 x = 3$$

4.

$$2 \sin^2 x + \cos x - 2 = 0$$

5.

$$\sec^2 x - 3 \tan x - 1 = 0$$

6.

$$\cos(2x) = \sin x$$

Give all real solutions

7.

$$\sin x = \frac{2}{5}$$

8.

$$\cos(2x) = \cos(5x)$$

9.

$$\sin(3x) = \cos x$$

10.

$$3 \csc^2 x - 4 \cot^2 x = 2$$

11.

$$\tan(2x) = -\sqrt{3}$$

12.

$$\sin^{-1}(2 \cos x) = -\frac{\pi}{6}$$